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Candidate Number

HONG KONG DIPLOMA OF SECONDARY
EDUCATION EXAMINATION BYTOPIC

MATHEMATICS Extended Part

Module 2 (Algebra and Calculus)

Question-Answer Book

Time allowed : 2 $\frac{1}{2}$ hours

This paper must be answered in English

INSTRUCTIONS

- (1) After the announcement of the start of the examination, you should first write your Candidate Number in the space provided on Page 1 and stick barcode labels in the spaces provided on Pages 1.
- (2) This paper consists of TWO sections, A and B.
- (3) This paper carried 100 marks.
- (4) Attempt ALL questions in this paper. Write your answers in the spaces provided in this Question-Answer Book. Do not write in the margins. Answers written in the margins will not be marked.
- (5) Graph paper and supplementary answer sheets will be supplied on request. Write your Candidate Number, mark the question number box and stick a barcode label on each sheet, and fasten them with string INSIDE this book.
- (6) Unless otherwise specified, all working must be clearly shown.
- (7) Unless otherwise specified, numerical answers must be exact.
- (8) The diagrams in this paper are not necessarily drawn to scale.
- (9) No extra time will be given to candidates for sticking on the barcode labels or filling in the question number boxes after the 'Time is up' announcement.

FORMULAS FOR REFERENCE

$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$	$\sin A + \sin B = 2 \sin \frac{A+B}{2} \cos \frac{A-B}{2}$
$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$	$\sin A - \sin B = 2 \cos \frac{A+B}{2} \sin \frac{A-B}{2}$
$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$	$\cos A + \cos B = 2 \cos \frac{A+B}{2} \cos \frac{A-B}{2}$
$2 \sin A \cos B = \sin(A + B) + \sin(A - B)$	$\cos A - \cos B = -2 \sin \frac{A+B}{2} \sin \frac{A-B}{2}$
$2 \cos A \cos B = \cos(A + B) + \cos(A - B)$	
$2 \sin A \sin B = \cos(A - B) - \cos(A + B)$	

SECTION A (63 marks)

1. (a) Prove by mathematical induction that $\sum_{k=1}^n \frac{k(k-1)}{2} = \frac{(n-1)n(n+1)}{6}$ for all positive integers n . (4 marks)

- (b) Evaluate $\sum_{k=1}^{10} \frac{k(k-1)}{2}$. (2 marks)

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2. (a) Prove by mathematical induction that $\sum_{i=1}^n \frac{1}{i^2 + i} = \frac{n}{n+1}$ for all positive integers n . (4 marks)

- (b) Evaluate $\sum_{i=1}^{99} \frac{1}{i^2 + i}$. (2 marks)

3. Let $\{a_n\}$ and $\{b_n\}$ be two sequences of positive integers. It is given that for all positive integers n ,

$$a_n = \sum_{k=1}^n (2^k + 1)$$

and $\{b_n\}$ satisfies the recurrence relation $b_1 = 3$ and $b_{n+1} = \frac{1}{2}b_n + a_n$ for $n \geq 1$.

(a) Prove by mathematical induction that for all positive integers n ,

$$a_n = 2^{n+1} + n - 2.$$

(4 marks)

(b) Prove by mathematical induction that for all positive integers n ,

$$b_n = \frac{1}{3} \left(2^{n+2} + 6n - 24 + \frac{19}{2^{n-1}} \right).$$

(4 marks)

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4. (a) Prove by mathematical induction that $\sum_{r=1}^n \frac{r}{r^4 + r^2 + 1} = \frac{n(n+1)}{2(n^2 + n + 1)}$ for all positive integers n . (4 marks)

(b) Evaluate $\sum_{r=33}^{64} \frac{r}{r^4 + r^2 + 1}$. (2 marks)

5. (a) Prove by mathematical induction that $\sum_{i=1}^n \left(\frac{1}{2}\right)^{-i} \cdot i^3 = 2^{n+1}(n^3 - 3n^2 + 9n - 13) + 26$ for all positive integers n . (4 marks)

(b) Evaluate $\sum_{i=8}^{13} \left(\frac{1}{2}\right)^{-i} \cdot i^3$. (2 marks)

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6. (a) Prove by mathematical induction that $\sum_{i=1}^n \frac{3i^4 + i^6}{(-1)^i} = \frac{(-1)^n}{2} n^3 (n+1)^3$ for all positive integers n . (4 marks)

(b) Evaluate $\sum_{i=9}^{16} \frac{3i^4 + i^6}{(-1)^{i+1}}$. (2 marks)

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7. (a) Prove by mathematical induction that $\sum_{n=1}^k \frac{1}{n(n+1)(n+2)} = \frac{k(k+3)}{4(k+1)(k+2)}$ for all positive integers k . (4 marks)
- (b) Using (a), prove that $\sum_{n=1}^k \frac{2n+k}{n(n+1)(n+2)} = \frac{k+4}{4} - \frac{5k+4}{2(k+1)(k+2)}$ for all positive integers k . (4 marks)
- (c) Evaluate $\sum_{n=6}^{18} \frac{2n+21}{(n+1)(n+2)(n+3)}$. (2 marks)

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8. (a) Prove by mathematical induction that $\sum_{k=1}^n \left(\frac{1+\sqrt{5}}{2} \right)^{k-1} \left(k \left(\frac{1+\sqrt{5}}{2} \right) - k + 1 \right) = n \left(\frac{1+\sqrt{5}}{2} \right)^n$ for all positive integers n . (4 marks)

(b) Evaluate $\sum_{k=1}^{10^{10}} \left(\frac{1+\sqrt{5}}{2}\right)^{(10^{10})} - \left(\frac{1+\sqrt{5}}{2}\right)^{k-3} \left(\left(\frac{1+\sqrt{5}}{2}\right) + 1\right) \left(k \left(\frac{1+\sqrt{5}}{2}\right) - k + 1\right)$.
(2 marks)

9. Let $\{u_n\}$ be a sequence whose first few terms are defined by $u_1 = 1$, $u_2 = 5$, and for all positive integers n , satisfies the following recurrence relation:

$$u_{n+2} = 5u_{n+1} - 6u_n$$

(a) Find the values of u_3 and u_4 . (2 marks)

(b) Prove by mathematical induction that for all positive integers n ,

$$u_n = 3^n - 2^n.$$

(4 marks)

(c) Let $S_n = \sum_{r=1}^n u_r$.

(i) By considering the result of (b), prove that $S_n = \frac{3^{n+1} - 2^{n+2} + 1}{2}$. (1 mark)

(ii) Prove by mathematical induction that for all positive integers n , $3^{n+1} - 2^{n+2} + 1$ is divisible by 2. (4 marks)

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SECTION B (37 marks)

10. (a) Prove by mathematical induction that $\sum_{k=1}^n \frac{1}{(k+1)\sqrt{k} + k\sqrt{k+1}} = 1 - \frac{1}{\sqrt{n+1}}$ for all positive integers n . (4 marks)

- (b) Evaluate $\sum_{k=1225}^{2024} \frac{1}{(k+1)\sqrt{k} + k\sqrt{k+1}}$. (2 marks)

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11. (a) Prove by mathematical induction that $\sum_{k=1}^n \tan k\theta \tan(k-1)\theta = \frac{\tan n\theta}{\tan \theta} - n$ for all positive integers n . (4 marks)

(b) Solve $\sum_{k=1}^6 \tan(k+6)\theta \tan(k+5)\theta = -6$, where $0 < \theta < \frac{\pi}{2}$. (3 marks)

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12. (a) Prove by mathematical induction that for all positive integers n ,

$$\sum_{k=1}^n \frac{\sin 2k\theta}{\cos(2k-1)\theta \cos(2k+1)\theta} = \frac{\sec(2n+1)\theta - \sec \theta}{2 \sin \theta}.$$

(4 marks)

(b) Evaluate $\sum_{k=1}^{256} \frac{\sin \frac{k\pi}{8}}{\cos \frac{(2k-1)\pi}{16} \cos \frac{(2k+1)\pi}{16}}.$

(2 marks)

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13. Let $B = \begin{pmatrix} 0 & 3 \\ 0 & 0 \end{pmatrix}$ and $A = 4I + B$, where I is the 2×2 identity matrix.

- (a) Find B^2 . (1 mark)
- (b) Prove by mathematical induction that $A^n = 4^n I + n4^{n-1}B$ for all positive integers n . (4 marks)

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14. Let $A = \begin{pmatrix} 2 & 0 & 0 \\ 2 & 2 & 0 \\ 0 & 2 & 2 \end{pmatrix}$.

Prove by mathematical induction that $A^n = \begin{pmatrix} 2^n & 0 & 0 \\ n2^n & 2^n & 0 \\ n(n-1)2^{n-1} & n2^n & 2^n \end{pmatrix}$ for all positive integers n . (4 marks)

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